

Problem-Solution Fit

Date	01.10.2026
Team ID	6ab295b91266c775575a6dce
Project Name	CRM APPLICATION FOR THE CHAI SHOP & CO
Maximum Marks	5 Marks

Problem-Solution Fit Canvas: Use the framework below to ensure your proposed solution accurately addresses the root causes behaviors and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

Box 1 CUSTOMER SEGMENT(S) [CS]: - Regular Chai Lovers - Office Workers - College Students - Age 18–!local daily tea/coffee & snacks - Chai Shop Owner - Staff/Cashiers	Box 6 CUSTOMER LIMITATIONS EG BUDGET / DEVICES [CL]: - Budget low: small chai shop cannot afford expensive CRM - Devices: basic Android phones - Limited internet during peak - Staff low tech knowledge — only Tamil/English - Cash payment preferred over UPI	Box 5 AVAILABLE SOLUTIONS PROS & CONS [AS]: - Manual Register — Pros: cheap - Cons: lost records, no analytics - Expensive CRMs (Zoho/Salesforce) - Pros: featurrs · Cons: costly, complex, not Tamil - POS-only machines — Pros: billing, no insights - Our solution: Affordable tailor-made chai shop Tamil/English UI · loyalty + inventory
Box 2 PROBLEMS / PAINS + ITS FREQUENCY [PR]: - No customer records — daily - No loyalty tracking — always - Manual billing slow during peak hours — daily - Inventory stockout — weekly - No feedback system - Customer churn due to no offers - Frequency: Daily rush 7–10am, 5–8pm	Box 9 PROBLEM ROOT / CAUSE [RC]: - Root cause: Manual paper-based system, no digital records - No integrated POS + inventory + CRM - No customer database to track preferences - No automation for loyalty/stock alerts - Lack of affordable technical solution for small chai shops	Box 10 YOUR SOLUTION [SL] CRM APPLICATION FOR THE CHAI SHOP & CO - React.js + Node.js + MongoDB Atlas web app - Fast billing <2 sec - OTP login, role-based: Owner / Staff / Customer - Loyalty: 1 point per Rs10 + 10% off 5th order - Inventory alert: <5kg notification - Bazormay: UPI / Cash payment - Fast2SMS + WhatsApp Business API notifications - Sales dashboard & analytics - Tamil/English UI, mobile responsive - Deployed on Docker/Kubernetes, low cost
Box 3 TRIGGERS TO ACT [TR]: - Customer asks for loyalty card - Owner loses regular customer - Peak hour long quene complaints - Stock finishes suddenly during rush - Competitor chai shop opens nearby with offers - Sales drop noticed monthly	Box 8 CHANNELS OF BEHAVIOR [CH] ONLINE: - Direct walk-in counter - Loyalty card referral discount - Phone call for bulk office orders	
Box 4 EMOTIONS BEFORE / AFTER [EM]:		
BEFORE: - Frustrated — waiting in long queue - Disappointed — no loyalty benefits - Owner stressed — manual work, inventory confusion		AFTER: - Happy — quick billing <2 sec - Satisfied — with loyalty points & rewards - Owner relaxed — automated reports & WhatsApp offers

Project Initialization and Planning Phase

Date	01 October 2026
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Project Proposal (Proposed Solution) report

The proposal report aims to transform traditional chai shop management using modern MERN stack CRM, boosting efficiency and customer retention. It tackles manual billing, inventory, loyalty inefficiencies, promising better operations, reduced stockouts, and happier customers with personalized offers. Key features include fast billing under 2 sec, auto loyalty points, inventory alerts, and WhatsApp notifications.

Project Overview	
Objective	The primary objective is to revolutionize the chai shop customer management process by implementing a cloud-based CRM application, ensuring faster billing, automated loyalty rewards, real-time inventory tracking, and enhanced customer satisfaction with Tamil/English UI.
Scope	The project comprehensively assesses and enhances the chai shop operation process, covering customer management, order billing, inventory management, loyalty program, sales analytics, incorporating MERN stack for a more robust and efficient system for small tea shops.
Problem Statement	
Description	Addressing inaccuracies and inefficiencies in the current manual chai shop system — no customer records, no loyalty tracking, slow billing during rush hours, frequent inventory stockouts, no feedback mechanism adversely affects operational efficiency and customer retention.
Impact	Solving these issues will result in improved operational efficiency, reduced inventory wastage, increased repeat customers via loyalty points (1 per Rs10 and 10% off on 5th order), and an overall enhancement in sales and customer satisfaction contributing to chai shop growth and profitability.
Proposed Solution	
Approach	Employing MERN stack — React.js, Node.js, Express, MongoDB Atlas — to create a dynamic and adaptable CRM application with role-based access for Owner/Staff/Customer, integrating Razorpay (UPI/Cash), Fast2SMS OTP, WhatsApp Business API, Google Maps for delivery, with mobile responsive design.
Key Features	• Fast billing under 2 seconds with auto inventory deduction • Automated loyalty system: 1 point per Rs10 spent, stock alert below 5kg • Customer management with OTP login, feedback ratings, referral rewards • Sales dashboard: daily/monthly analytics • WhatsApp Business API for offer alerts and order ready notifications • Tamil/English bilingual UI • Real-time decision-making for quicker billing and inventory checks • Continuous learning to adapt to evolving customer preferences via analytics
Team	

	- Real-time decision-making for quicker billing and inventory checks. - Continuous learning to adapt to evolving customer preferences via analytics.
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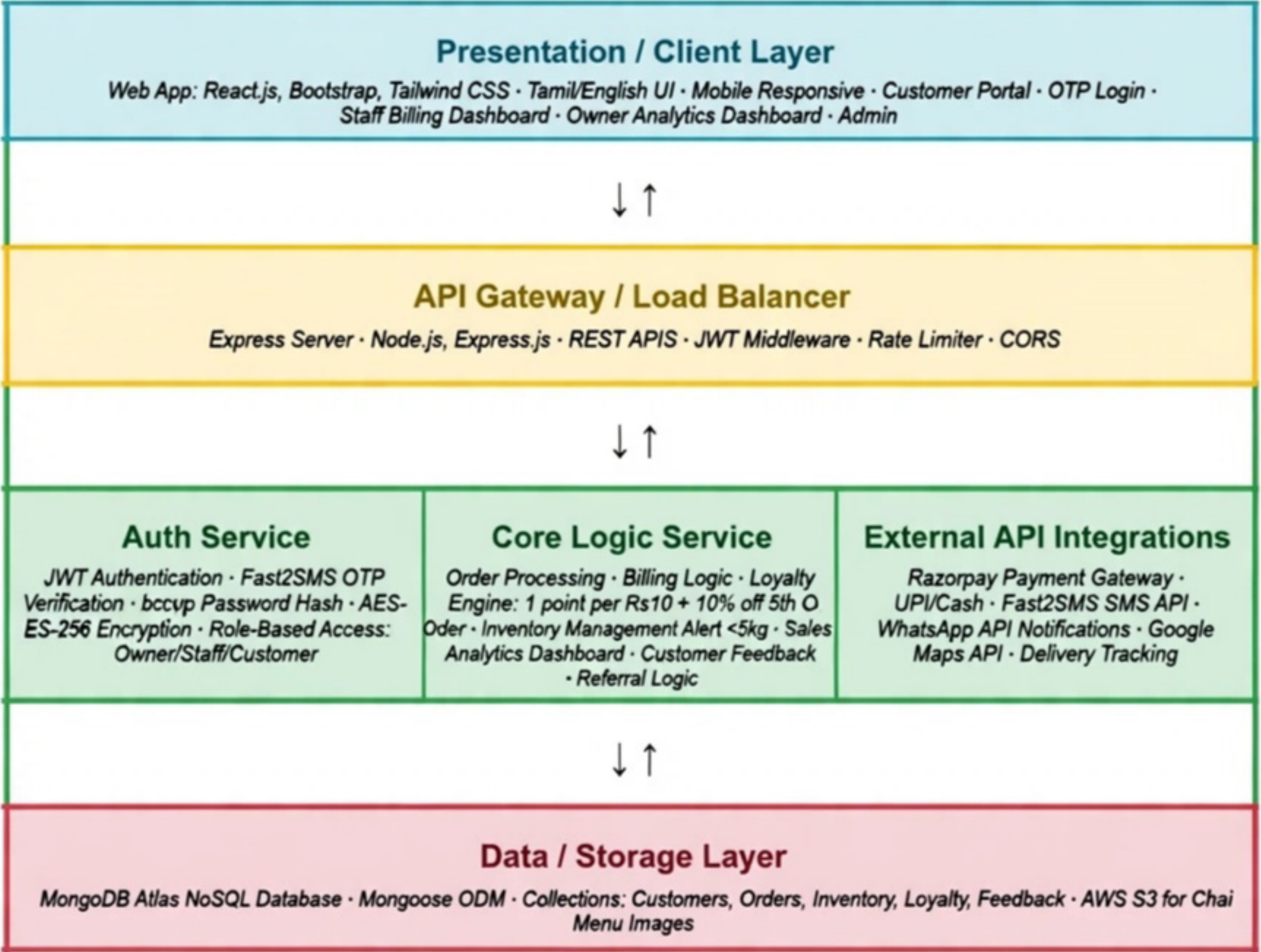
Resource Requirements

Specification/Allocation

Computing Resources	Description	
	CPU/GPU specifications, number of cores	Intel i5 11th Gen, 4 cores; 8GB RAM, 256GB SSD development laptop; T4 GPU not needed
Memory	RAM specifications	8 GB minimum; 16 GB recommended for MongoDB & React dev server
Storage	Disk space for data, models, and logs	500 GB SSD for code, database backups, customer images, chai menu photos
Software		
Frameworks	Web frameworks	MERN Stack: React.js 18 (Frontend — Bootstrap + Tailwind); Node.js 20, Express.js 4 (Backend)
Libraries	Additional libraries	React-Router-DOM, Axios, Mongoose, JWT (jsonwebtoken), bcrypt, Multer, Nodemailer, Chart.js for analytics, qrcode
Development Environment	IDE / Tools	
	VS Code IDE; Git & GitHub for version control; Postman for API testing; Chrome DevTools; MongoDB Compass	50,000 orders (CSV/JSON); Inventory — chai, tea, coffee, snacks
Data		
Source, size, format	MongoDB Atlas collections: Customers — 10,000 records; Orders — 50,000 orders (CSV/JSON);	snacks items (100 items); Feedback & ratings; Loyalty points transactions; Sales reports — daily/monthly analytics data

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Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Handles user interface for customers, staff, owner responsive billing, ordering, loyalty tracking UI	React.js 18 · Bootstrap 5 · Tailwind CSS · HTML5 · CSS3 · JavaScript · Axios
API Gateway	Manages API requests, routing, authentication middleware, load balancing	Node.js 20 · Express.js 4 · JWT · CORS · dumen · Rate Limiter
Microservices	Contains 3 services: Auth Service handles OTP login, encryption, role access; Core Logic handles billing, loyalty, inventory, analytics; External API handles Razorpay, SMS, WhatsApp, Maps integration	JWT · Fast2SMS SDK · Razorpay SDK · WhatsApp Business API · Google Maps API · bccrvp · AEs-256
Database	Stores all chai shop data securely with backup	MongoDB Atlas · Mongoose ODM · MongoDB Compass · AWS S3